THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 3: **The Atomic Foundations of Matter** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The formula of iron(III) oxide is: 1

(a) Fe_2O **(b)** FeO **(c)** Fe_2O_3 **(d)** Fe_3O_2

2. Which compound has a triple bond? 1

(a) Cl_2 **(b)** N_2 **(c)** H_2 **(d)** O_2

3. In CO2, C : O = 3 : 8. The mass of CO2 formed from 6 g of carbon is: 1

(a) 16 g **(b)** 22 g **(c)** 14 g **(d)** 11 g

4. The formula of copper(II) sulphate is: 1

(a) Cu_2SO_4 **(b)** $CuSO_3$ **(c)** $CuSO_4$ **(d)** $Cu(SO_4)_2$

5. The number of shared pairs in CO_2 (total) is: 1

(a) 1 **(b)** 2 **(c)** 4 **(d)** 3

6. The ratio of the masses of carbon to oxygen in CO_2 (C = 12, O = 16) is: 1

(a) 8 : 3 **(b)** 3 : 4 **(c)** 1 : 2 **(d)** 3 : 8

7. The formula unit mass of Na_2CO_3 is (Na = 23, C = 12, O = 16): 1

(a) 122 u **(b)** 100 u **(c)** 84 u **(d)** 106 u

8. The number of electrons transferred in the formation of MgO is: 1

(a) 3 **(b)** 4 **(c)** 1 **(d)** 2

9. 46 g of sodium reacts with 71 g of chlorine to form: 1

(a) 46 g NaCl **(b)** 71 g NaCl **(c)** 58.5 g NaCl **(d)** 117 g NaCl

10. The formula unit mass of $CaCl_2$ is (Ca = 40, Cl = 35.5): 1

(a) 111 u **(b)** 151 u **(c)** 95.5 u **(d)** 75.5 u

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. How many atoms are present in one formula unit of $Ca(OH)_2$? 2

12. Explain why NaCl is a solid at room temperature while HCl is a gas. 2

13. Find the formula unit mass of $Mg(OH)_2$ (Mg = 24, O = 16, H = 1). 2

14. State two limitations of Dalton's atomic theory. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how the formation of magnesium chloride involves transfer of electrons. 3

16. Explain how to write the formula of aluminium sulphate using the criss-cross method. 3

17. Write the electronic arrangements and explain the ion formed by 3
(a) Na
(b) F.

OR

Draw the electron-sharing diagram in words for the water molecule. 3

18. What is meant by formula unit mass? Calculate it for $CaCO_3$. 3

OR

Explain why diamond, though covalent, has a very high melting point. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain covalent bonding in CH_4, NH_3, H_2O and CO_2 with the number of bonds formed by each central atom. 5

OR

Discuss the nature of ionic lattices, the arrangement of Na+ and Cl- ions in NaCl, and the properties following from the structure. 5

20. Differentiate between ionic and covalent compounds on five properties with examples and explain the reasons for any two. 5

OR

Calculate the formula unit masses and the percentage of metal in MgO, CaO and Fe_2O_3 and explain which has the highest metal percentage. 5

END OF QUESTION PAPER — Total: 40 Marks

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